

Proposal

Project title	LearnHub – Web-Based Learning Management System
Module	CT050-3-2-WAPP Web Applications (group assignment)
Institution	Asia Pacific University of Technology & Innovation (APU)
Team	Member 1 (TP000000), Member 2 (TP000000), Member 3 (TP000000), Member 4 (TP000000)
Technology	ASP.NET Core MVC on .NET 10, C#, Razor, Bootstrap 5.3, Entity Framework Core 10 with SQLite, ASP.NET Core Identity

1. Background and problem

Computing students learn from many scattered sources: lecture slides, videos, blog posts and cheat sheets. Three problems follow:

1. **No structure.** Material has no clear order, so beginners do not know what to study next or how far they are from finishing a topic.
2. **No feedback.** Reading and watching give no evidence of understanding. Students need short quizzes with explanations, not only a score.
3. **No single place to manage content.** Lecturers and teaching assistants need one tool to publish courses, lessons, files and quizzes, see who is learning and remove outdated material, without editing web pages by hand.

2. Proposed solution

LearnHub is a web application in which:

- **guests** explore a catalogue of computing courses, read course details and try free preview lessons;
- **students** register, enrol in courses, work through lessons in order, mark them complete, take quizzes with explained answers and follow their progress on a personal dashboard;
- **administrators** manage courses, categories, lessons, uploads, quizzes, enrolments, users and contact messages in a protected admin area with live statistics.

The interface uses one idea throughout: LearnHub is an **interchange**. Each subject category is a coloured transit line, a course is a route, lessons are stops, and progress is how far along the route a student has travelled.

3. Mission statement

LearnHub gives every computing student a clear route through each subject, with honest feedback at every stop, and gives administrators one simple place to keep that route up to date.

4. Objectives

#	Objective (SMART)	Measure of success
O1	Deliver a public catalogue in which guests can search, filter and open published courses within the first four weeks.	Search, category and difficulty filters and course details pages work, and preview lessons open without an account.
O2	Let a student register, enrol, complete lessons and take a quiz in one continuous journey.	An automated end-to-end test covers register → enrol → lesson → quiz → result.
O3	Show each student meaningful progress data on a dashboard built from real database records.	Dashboard shows enrolled, completed and available courses, lessons completed, quiz results, recent activity and recommendations.
O4	Provide administrators with full create, read, update and delete management of all learning content, enrolments and users.	Every admin entity has working list, create, edit and delete pages, tested through the real forms.
O5	Protect the application against the common web risks in the OWASP Top 10.	Security review completed; automated tests for access control, CSRF, XSS, uploads and logout pass.
O6	Make every page usable on phones, tablets and desktops and accessible to keyboard and screen-reader users.	Browser tests pass at three screen sizes with no WCAG 2.2 A/AA violations reported by axe-core.
O7	Ship the application as a Docker image to Railway, publish the static presentation site on Firebase Hosting, and run automated tests before every delivery.	The three CI jobs (<code>build-and-test</code> , <code>container</code> , <code>e2e</code>) pass; the presentation site is live and links to the application; the Railway deployment files and commands are complete and ready to run.

5. Target audience

Persona 1 – Nurul, first-year computing student

Age and situation	19, Diploma in IT, lives in a hostel, studies between classes
Devices	Mid-range Android phone on campus Wi-Fi; shared laptop in the library
Goals	Understand programming and databases fundamentals before assessments; know what to study next
Frustrations	Long videos without structure; not knowing whether she really understood a topic
What LearnHub offers	Short lessons in a fixed order, a Resume button on the dashboard, quizzes that explain every answer, and a layout that works on a phone

Persona 2 – Daniel, career switcher

Age and situation	31, works full time in logistics, learning web development in the evenings
Devices	Personal laptop; phone during commutes

Goals	Build practical skills in HTML, CSS, JavaScript, ASP.NET Core and security; see steady progress
Frustrations	Limited time; losing track of where he stopped; content that assumes prior knowledge
What LearnHub offers	Difficulty levels and durations on every course, progress bars, "Continue learning" with the next lesson, and PDF cheat sheets for quick revision

Persona 3 – Ms. Priya, lecturer and course administrator

Age and situation	42, lecturer responsible for networking and cloud modules
Devices	Office desktop; tablet in meetings
Goals	Publish and update course material quickly; check enrolments and quiz performance; keep content accurate
Frustrations	Complex tools; accidentally deleting material; no overview of learner activity
What LearnHub offers	A focused admin area with a dashboard, publish and unpublish without deleting, confirmation pages that show the impact of a deletion, validated uploads and quiz result reports

6. Scope

In scope

Role	Features
Guest	Home, About, Contact form, Privacy; course catalogue with search, filters, sorting and pagination; course details; free preview lessons; registration and login
Student	Dashboard; profile and password change; enrol and leave courses; My Courses; lessons (article, video, PDF, image, link); mark complete; progress; quizzes with instant marking and explanations; result history; logout
Administrator	Dashboard with statistics; courses (with cover images and publishing); categories; learning resources (with uploads); quizzes, questions and answer options; enrolments; quiz results; users (roles, deactivation, deletion); contact messages
Platform	ASP.NET Core Identity with Admin and Student roles; a single SQLite database with EF Core code-first migrations and demo data; security controls; responsive and accessible design; automated tests; CI; the application packaged as a Docker image for Railway; a Firebase Hosting presentation site

Out of scope for version 1.0

AI tutor, live video classes, chat, payments, certificates, email delivery (email confirmation and password reset by email), an instructor self-service role, discussion forums, native mobile apps and multiple interface languages.

7. Requirements summary

The complete, numbered list is in [REQUIREMENTS_CHECKLIST.md](#).

Group	IDs	Examples
Technology	TECH-01–11	.NET 10 MVC, EF Core, Identity, Razor, Bootstrap, JavaScript, multimedia
Guest	GUEST-01–10	Browse, search, filter, details, preview, register, login, contact
Student	STU-01–13	Dashboard, enrolment, lessons, progress, quizzes, results, profile
Administrator	ADM-01–11	Protected area, CRUD for all content, users, enrolments, results, messages
Database	DB-01–10	Keys, constraints, indexes, delete rules, migrations, seed data, ERD
Security and validation	AUTH, VAL, SEC	Password hashing, roles, CSRF, XSS, IDOR, overposting, upload safety
Quality	ERR, UI, TEST	Error pages, logging, responsive and accessible UI, automated and manual tests
Delivery	GIT, CI, DEP, HOST, DOC	Repository, workflows, Railway container deployment, Firebase Hosting site, documentation

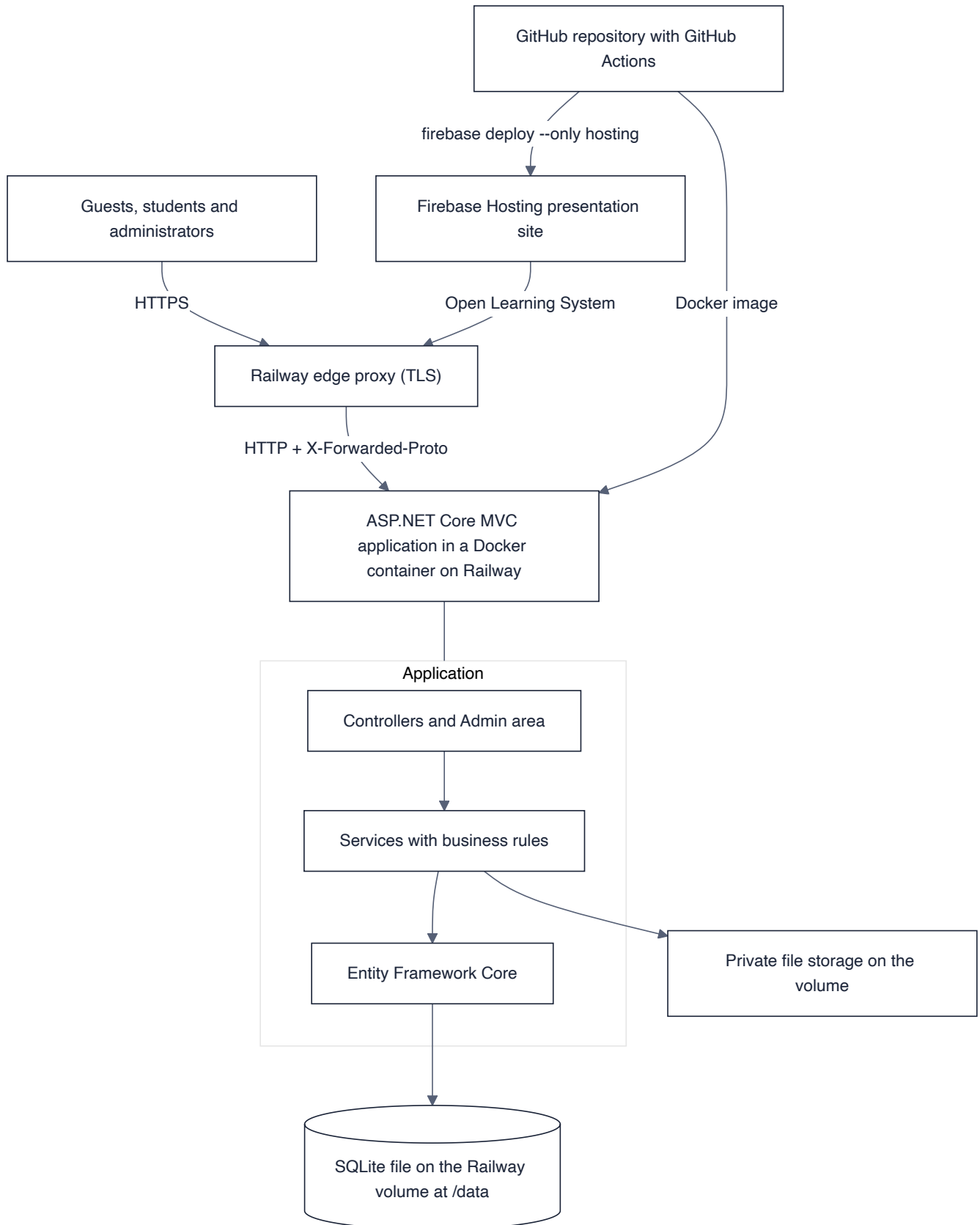
Non-functional requirements: pages respond within about one second once the container is warm; the interface meets WCAG 2.2 A/AA contrast and keyboard access; no secret is stored in the repository — the administrator password is supplied through `Seed__AdminPassword` in the environment and every other credential is generated per CI run; the application can be rebuilt from an empty database with migrations and seed data.

8. Proposed technology

Choice	Reason
ASP.NET Core MVC on .NET 10 (LTS)	Required .NET technology; long-term support until 2028; built-in model binding, validation, antiforgery and Razor views
C#	Strong typing and the language of the module
Razor views with HTML5 semantic markup	Server-rendered pages that work without JavaScript and stay accessible to assistive technology
Entity Framework Core 10 with SQLite	LINQ queries are parameterised by default, code-first migrations keep the schema in version control, and the whole database is one file that needs no server
SQLite on a Railway volume	One provider in development, testing and production, so no behaviour can differ between environments; the database file and the Data Protection key ring live on a mounted volume at <code>/data</code> , so a redeploy keeps every account and upload; nothing external has to be provisioned, explained or demonstrated

Choice	Reason
ASP.NET Core Identity	Proven password hashing, lockout, Admin and Student roles and cookie authentication instead of custom security code
Bootstrap 5.3 with a custom design system	Accessible, responsive components, restyled so the site has its own identity
Vanilla JavaScript and jQuery Validation Unobtrusive	Progressive enhancement; client validation generated from the same server rules
xUnit and Playwright	Fast automated tests of the real application, plus browser tests on several screen sizes
GitHub and GitHub Actions	Version control, pull requests and the three CI jobs that build, run the container and exercise the site in a browser
Railway (Docker image)	Runs the application itself plus its <code>/data</code> volume and its TLS edge proxy; a Dockerfile and <code>railway.json</code> describe the whole deployment
Firebase Hosting	Serves the static presentation site from <code>FirebaseLanding/</code> over HTTPS with its own caching rules, for free

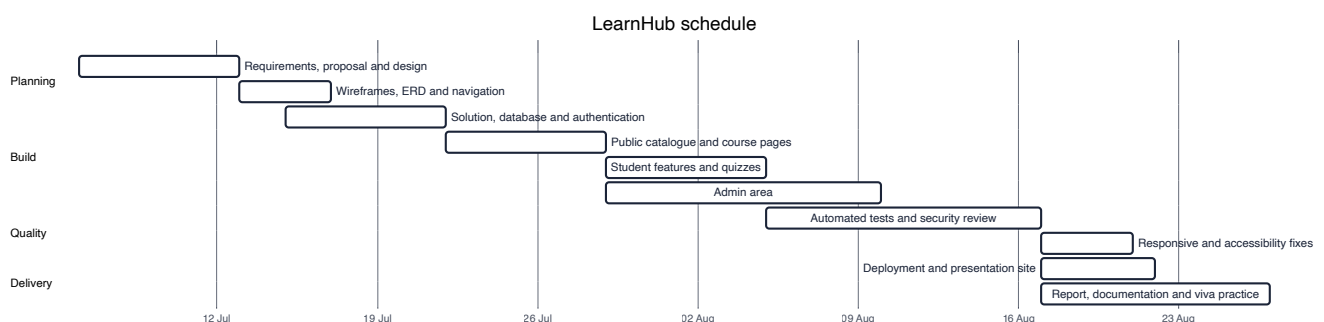
9. High-level architecture



10. Risks and mitigations

Risk	Likelihood	Impact	Mitigation
Team members have little disk space for the .NET SDK	High	Medium	Build, test and generate migrations in GitHub Actions; only the repository is needed locally
Passwords or connection strings leak into Git	Medium	High	User secrets locally; environment variables on Railway; <code>.env</code> is git-ignored and only <code>.env.example</code> with placeholders is committed; CI generates throw-away passwords and masks them
The SQLite file is lost on redeploy because the volume is missing	Medium	High	The database, the Data Protection key ring and uploads all live under <code>/data</code> ; CI mounts a volume in the same place and restarts the container to prove migrations are not re-applied
A single SQLite instance cannot serve many concurrent writers	Low	Medium	The deployment is deliberately single-instance and read-heavy, which suits one file; the service layer is provider-agnostic should a server database ever be needed
Scope creep (payments, chat, certificates)	Medium	Medium	Fixed out-of-scope list; new ideas recorded as future enhancements
Security weaknesses in custom code	Medium	High	Use framework features (Identity, antiforgery, Razor encoding); security review; attack-style tests
Uneven contribution or merge conflicts	Medium	Medium	Clear ownership per area, short-lived branches, pull requests with CI checks
Cloud free-tier limits or cold starts during the demonstration	Medium	Low	The health check runs before the service accepts traffic; warm the site before presenting; screenshots and a local container run as backup

11. Schedule



12. Deliverables

1. Source code in the GitHub repository with a README.
2. The application packaged as a Docker image for Railway (`Dockerfile` , `railway.json` , `.env.example`) and the presentation site live on Firebase Hosting at <https://learnhub-wapp.web.app>.

3. Automated tests and the three CI jobs in `.github/workflows/ci.yml`.
4. Documentation: this proposal, the final report content, ERD, use cases, flowcharts, wireframes, navigation structure, testing and security documents, deployment guide, team responsibilities, Git workflow and viva preparation.